WORKSHEET

Chapter 8: Discrete and continuous random variables

1 Data that can be counted is called:

A Variable data

B Continuous data

C Ordinal data

D Discrete data

E Interval data

2 Data for which data values are measured is called:

A Ordinal data

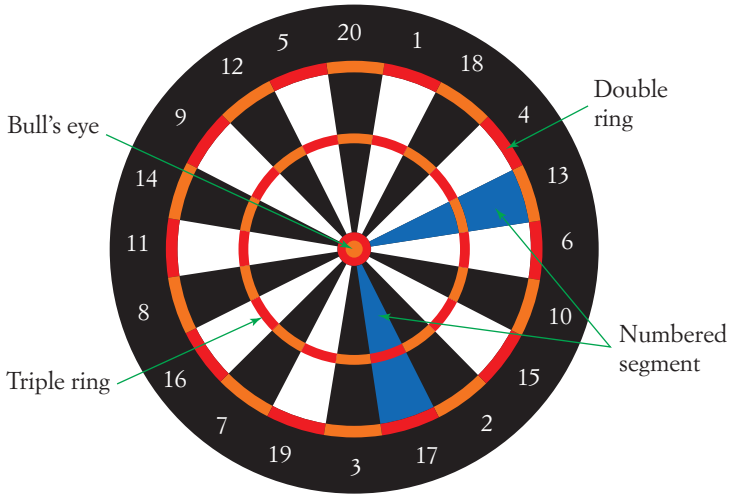
B Variable data

C Continuous data

D Discrete data

E Interval data

3 Consider the situation where a novice dart player throws a dart at a dart board. For this novice player, the dart is equally likely to land on any spot on the board on any throw.



The score for any dart thrown is calculated as follows:
Bull's eye – 50 points
Ring outside the bull's eye – 25 points
Numbered segment – The same number of points as the segment number
Double ring – Double the points of the segment number
Triple ring – Triple the points of the segment number

Classify each of the following variables as either discrete or continuous and random or non-random:

a the numbered segment (1, 2, 3, …, 20) on which the dart lands

b the distance the dart finishes from the bull's eye

c the time taken for the dart to reach the dart board

d the total score the player earns from throwing three darts.

4 A professional darts player replaces the novice at the dart board. Classify each of the following variables as either discrete or continuous and random or non-random.

a the numbered segment (1, 2, 3, …, 20) on which the dart lands

b the distance the dart finishes from the bull's eye

c the time taken for the dart to reach the dart board

d the total score the player earns from throwing three darts.

5 Consider the situation where a pair of dice is rolled. Classify each of the following variables as either discrete or continuous and random or non-random:

a the sum of the faces of the dice

b the time taken for the dice to come to rest

c the number of times a six occurs in 10 rolls

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d the distance between the dice after they are rolled

e the number of times an odd total occurs in 20 rolls.

6 Classify these variables as either discrete or continuous and random or non-random:

a the number of heads when four coins are tossed

b the number of pieces of fruit eaten by the students in a class

c the heights of a group of students

d the lifetime of a battery

e the number of goals scored each week by a netball team

f the number of children in an Australian family

g the number of loaves of bread bought each week by a family

h the number of pets owned by students in a Year 12 class

i the number strokes taken by a professional golfer to complete 18 holes of golf

j the length of time the sun shines in a day

k the number of people who die from cancer each year in Australia

l the amount of rainfall in each month of the year

m the time spent by students doing homework

n the marks scored in a class test

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o the stopping distances of cars doing 60 km/h

p the pulse rates of a group of athletes at rest

q Australia's annual road toll

r the water intake of people during a heat wave.

Answers

1 D

2 C

3 **a** Discrete, random
b Continuous, random
c Continuous, random
d Discrete, random

4 **a** Discrete, non-random
b Continuous, non-random
c Continuous, random
d Discrete, non-random

5 **a** Discrete, random
b Continuous, random
c Discrete, random
d Discrete, random
e Discrete, random

6 **a** Discrete, random
b Discrete, random
c Continuous, random
d Continuous, random
e Discrete, random
f Discrete, random
g Discrete, random
h Discrete, random
i Discrete, non-random
j Continuous, random
k Discrete, random
l Continuous, random
m Continuous, random
n Discrete, random
o Continuous, random
p Discrete, non-random
q Discrete, random
r Continuous, non-random

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